

Answers learning task 5

2026 – 2027

Exercise 4.1: Suppose that $f(x) = e^{-2x}$ for $0 < x$. Determine the following:

- (a) $P(1 < X)$
- (b) $P(1 < X < 2.5)$
- (c) $P(X = 3)$
- (d) $P(X < 4)$
- (e) $P(3 \leq X)$
- (f) Determine x such that $P(x < X) = 0.20$.
- (g) Determine x such that $P(X \leq x) = 0.10$.

Answer

- (a) 0.1353
- (b) 0.1286
- (c) 0
- (d) 0.9997
- (e) 0.0025
- (f) 0.8047
- (g) 0.0527

Exercise 4.7: Suppose that $f(x) = 2x^2$ for $-1 < x < 1$. Determine the following:

- (a) $P(0 < X)$
- (b) $P(0.5 < X)$
- (c) $P(-0.5 \leq X \leq 0.5)$
- (d) $P(X < -2)$
- (e) $P(X < 0 \text{ or } X > -0.5)$
- (f) x such that $P(x < X) = 0.05$.

Answer

- (a) 0.5
- (b) 0.4375
- (c) 0.125
- (d) 0
- (e) 1
- (f) 0.9655

Exercise 4.11: The probability density function of the length of a metal rod is $f(x) = 2$ for $2.3 < x < 2.9$ meters.

- (a) If the specifications for this process are from 2.25 to 2.75 meters, what proportion of rods fail to meet the specifications?
- (b) Assume that the probability density function is $f(x) = 2$ for an interval of length 0.5 meters. Over what value should the density be centered to achieve the greatest proportion of rods within specifications?

Answer

- (a) 0.1
- (b) 2.5, thus $f(x) = 2$ for $2.25 < x < 2.75$

Exercise 4.13: Suppose that the cumulative distribution function of the random variable X

is

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.2x & 0 \leq x < 5 \\ 1 & 5 \leq x \end{cases}$$

Determine the following:

(a) $P(X < 2.8)$

(b) $P(X > 1.5)$

(c) $P(X < -2)$

(d) $P(X > 6)$

Answer

(a) 0.56

(b) 0.7

(c) 0

(d) 0

Exercise 4.15: Determine the cumulative distribution function given the following probability density function:

$$f(x) = e^{-2x} \quad \text{for } 0 < x$$

Answer

$$F(x) = \begin{cases} 0 & x \leq 0 \\ 1 - e^{-2x} & x \geq 0 \end{cases}$$

Exercise 4.21: Determine the cumulative distribution function for the following probability density function:

$$f(x) = \begin{cases} 2, & \text{for } 2.3 < x < 2.9 \\ 0, & \text{otherwise.} \end{cases}$$

Use the cumulative distribution function to determine the probability that a length exceeds 2.7 meters.

Answer

$$F(x) = \begin{cases} 0 & x \leq 2.3 \\ 2x - 4.6 & 2.3 \leq x \leq 2.8 \\ 1 & x \geq 2.8 \end{cases}$$
$$P(X > 2.7) = 0.2$$

Exercise 4.29: Suppose that $f(x) = 2x^2$ for $-1 < x < 1$. Determine the mean and variance of X .

Answer

$$E(X) = 0, \quad V(X) = 0.6$$

Exercise 4.37: Integration by parts is required. The probability density function of the diameter of a drilled hole in millimeters is $10e^{-10(x-5)}$ for $x > 5$ mm. Although the target diameter is 5 millimeters, vibrations, tool wear, and other nuisances produce diameters greater than 5 millimeters.

- (a) Determine the mean and variance of the diameter of the holes.
- (b) Determine the probability that a diameter exceeds 5.1 millimeters.

Answer

- (a) $E(X) = 5.1, \quad V(X) = 0.01$
- (b) 0.3679

Exercise 4.41: The thickness of a flange on an aircraft component is uniformly distributed between 0.90 and 1.05 millimeters.

- (a) Determine the cumulative distribution function of flange thickness.
- (b) Determine the proportion of flanges that exceeds 1.02 millimeters.
- (c) What thickness is exceeded by 90 % of the flanges?
- (d) Determine the mean and variance of flange thickness.

Answer

- (a)
$$F(x) = \frac{20}{3}x - 6 \quad \text{for } 0.9 \leq x \leq 1.05$$
- (b) 0.2
- (c) 0.915
- (d) $E(X) = 0.975, \quad V(X) = 0.001875$